Dog Emotion Classification

Using CNNs, Transfer Learning and Ensemble Learning

Deep Learning Image Classification Project

4-Class Dog Emotion Classification

Angry — Happy — Relaxed — Sad

Project Report

Deep Learning and Computer Vision

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Abstract

This project investigates automated dog emotion classification from images using convolutional neural networks and transfer learning. The objective is to classify dog images into four emotion categories: *Angry, Happy, Relaxed, and Sad*.

A custom CNN was first developed as a baseline model. Five ImageNet-pretrained CNN architectures—MobileNetV2, EfficientNetB0, ResNet50, Xception and InceptionV3—were subsequently evaluated using transfer learning and fine-tuning. Finally, the two strongest architectures, EfficientNetB0 and ResNet50, were combined using a feature-level ensemble architecture.

The dataset was divided into 70% training, 15% validation and 15% test data. The test set was kept untouched during model training and model selection.

The baseline CNN achieved a test accuracy of 61.95%, while all pretrained architectures achieved substantially higher performance. The final EfficientNetB0 + ResNet50 ensemble achieved the best test accuracy of 84.13%, with a macro recall of 84.03% and macro F1-score of 84.10%.

These results demonstrate the effectiveness of transfer learning and feature-level ensemble learning for visual dog emotion classification.

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1 Introduction

Understanding animal behaviour from visual information is an interesting problem at the intersection of computer vision and behavioural analysis. Dogs communicate their emotional state through facial expressions, posture, body orientation and other visual cues.

Manual interpretation of these cues can be subjective and may vary between observers. A computer vision system capable of learning patterns associated with different emotional states could therefore provide a useful automated classification framework.

The goal of this project is to develop a deep learning system that takes an image of a dog as input and predicts one of four emotion classes:

- Angry
- Happy
- Relaxed
- Sad

The project follows a progressive modelling strategy:

1. Establish a custom CNN baseline.
2. Evaluate multiple pretrained CNN architectures.
3. Fine-tune the strongest pretrained models.
4. Combine complementary models using feature-level ensemble learning.
5. Evaluate all models on an untouched test set.

2 Problem Statement

Given an image containing a dog, the objective is to automatically classify the dog's apparent emotional state into one of four classes:

$$Y \in \{\text{Angry}, \text{Happy}, \text{Relaxed}, \text{Sad}\}$$

This is formulated as a multiclass image classification problem.

The model receives an RGB image:

$$X \in R^{H\times W\times 3}$$

and produces a probability vector:

$$P(Y|X) = [p_1, p_2, p_3, p_4]$$

where the predicted class corresponds to the emotion having the highest predicted probability.

3 Project Objectives

The major objectives of the project are:

1. Develop a CNN baseline for four-class dog emotion classification.
2. Establish a robust 70–15–15 train–validation–test experimental setup.
3. Apply image augmentation to improve generalization.
4. Benchmark multiple ImageNet-pretrained CNN architectures.
5. Use transfer learning followed by fine-tuning.
6. Compare models using validation and final test performance.
7. Evaluate class-level performance using precision, recall and F1-score.
8. Investigate whether combining strong models improves classification performance.

4 Dataset

The project uses a labelled image dataset containing four dog emotion categories:

Table 1: Dog Emotion Classes

Class	Description
Angry	Images representing an angry or aggressive expression
Happy	Images representing a happy or positive expression
Relaxed	Images representing a calm or relaxed expression
Sad	Images representing a sad or subdued expression

The final evaluation contained 586 test images.
The class distribution of the test set was:

Table 2: Test Set Class Distribution

Emotion	Number of Images
Angry	137
Happy	156
Relaxed	139
Sad	154
Total	586

The class counts are relatively balanced, which reduces the risk that overall accuracy is dominated by a single majority class.

5 Train–Validation–Test Split

A fixed 70–15–15 split was used throughout the project.

Figure 1: Fixed 70–15–15 train–validation–test split

The training set was used to learn model parameters. The validation set was used for early stopping, learning-rate scheduling and model comparison.

The test set remained untouched until final evaluation.

This separation is important because repeatedly splitting the dataset inside every model-training function could lead to inconsistent experimental conditions and potentially introduce misleading comparisons.

Therefore, the same underlying split was maintained for all models.

6 Data Preprocessing and Augmentation

Images were resized according to the input requirements of each architecture.

The baseline CNN used:

$$160 \times 160 \times 3$$

The pretrained architectures used their respective standard input resolutions:

Table 3: Input Resolution of Pretrained Architectures

Architecture	**Input Size**
MobileNetV2	160×160
EfficientNetB0	224×224
ResNet50	224×224
Xception	224×224
InceptionV3	299×299

Data augmentation was applied during training:

- Random horizontal flipping
- Random rotation
- Random zoom

The purpose of augmentation was to expose the model to small variations in orientation and appearance and thereby improve generalization.

7 Baseline CNN

A custom CNN was developed as the baseline.

Its architecture consisted of four convolutional blocks followed by global average pooling and fully connected layers.

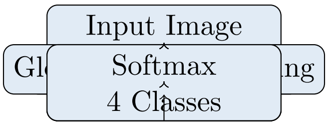

Figure 2: Architecture of the custom CNN baseline

Batch normalization, dropout and L2 regularization were used to improve optimization and reduce overfitting.

The baseline provides an important reference point for evaluating whether transfer learning actually provides a meaningful improvement.

8 Transfer Learning

Five pretrained CNN architectures were evaluated:

1. MobileNetV2
2. EfficientNetB0
3. ResNet50
4. Xception
5. InceptionV3

Each architecture was initialized with ImageNet-pretrained weights.

Initially, the pretrained convolutional backbone was frozen and a new classification head was trained.

The final classification head consisted of:

- Global Average Pooling
- Dropout
- Dense layer with 128 units
- Dropout
- Four-unit softmax output layer

9 Fine-Tuning

After training the classification head, the final 30 layers of each pretrained backbone were unfrozen.

The model was then trained using a smaller learning rate:

$$\eta = 10^{-5}$$

compared with:

$$\eta = 10^{-3}$$

during initial head training.

This two-stage approach allows the model to first adapt the newly introduced classification layers before making smaller adjustments to the pretrained visual representations.

10 Feature-Level Ensemble

EfficientNetB0 and ResNet50 were selected for ensemble learning because they were the strongest individual architectures based on their final test performance.

Instead of simply averaging predictions, the two networks were combined at the feature level.

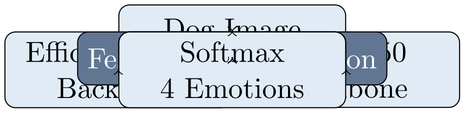

Figure 3: Feature-level EfficientNetB0 + ResNet50 ensemble

Let the feature representations produced by the two networks be:

$$f_1 = \text{EfficientNetB0}(X)$$

and

$$f_2 = \text{ResNet50}(X)$$

The combined representation is:

$$f = [f_1; f_2]$$

where $[\,;\,]$ denotes concatenation.

The combined representation is passed through additional dense layers before the final softmax classifier.

11 Evaluation Metrics

The models were evaluated using validation loss, validation accuracy, test loss and test accuracy.

For class-level evaluation, precision, recall and F1-score were also considered.

Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Recall is:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is:

$$F1 = 2\frac{Precision \cdot Recall}{Precision + Recall}$$

Macro F1 was used to treat all four emotion classes equally.

Angry-class recall was also specifically monitored because failing to detect Angry images represents an important class-level error.

12 Results

12.1 Validation and Test Performance

Table 4: Validation and Test Performance of All Models

Model	Val Loss	Val Acc.	Test Loss	Test Acc.
Ensemble	0.4590	0.8507	0.4955	0.8413
ResNet50	0.4405	0.8368	0.4736	0.8294
EfficientNetB0	0.4914	0.8385	0.5215	0.8276
Xception	0.5270	0.7951	0.5228	0.8072
InceptionV3	0.5969	0.7743	0.5996	0.7713
MobileNetV2	0.6880	0.7535	0.6629	0.7321
Baseline CNN	0.8858	0.6684	0.9961	0.6195

The final ensemble achieved the highest test accuracy of **84.13%**.

Among individual architectures, ResNet50 achieved the highest test accuracy of **82.94%**, closely followed by EfficientNetB0 at **82.76%**.

13 Model Accuracy Comparison

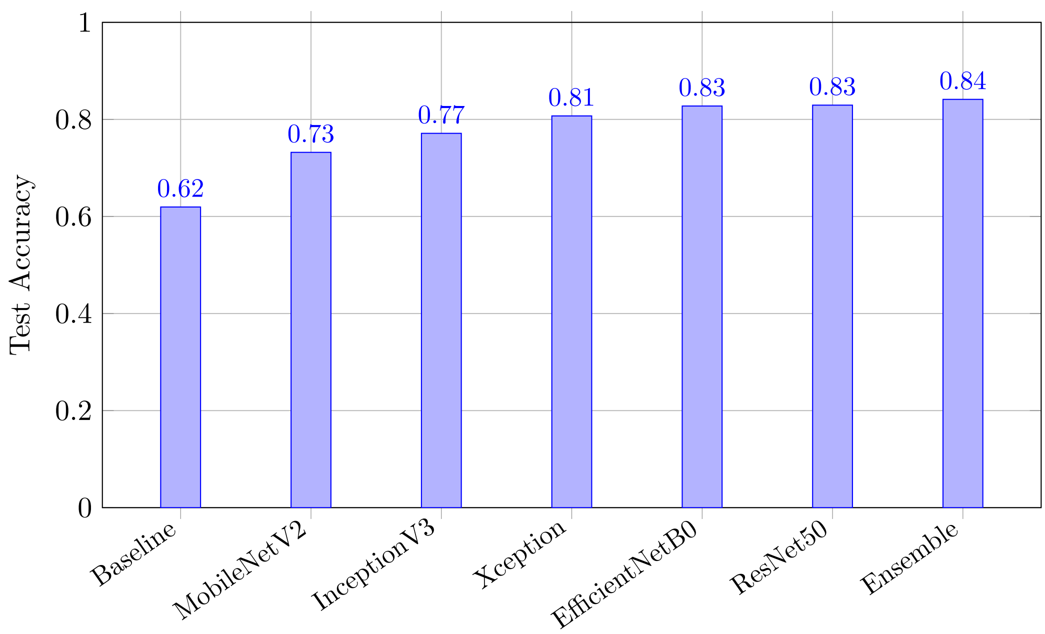

Figure 4: Test accuracy comparison across all models

The results show a clear improvement when moving from the custom CNN to pretrained architectures.

14 Class-Level Performance

Table 5: Final Test-Set Classification Metrics

Model	Accuracy	Macro Recall	Macro F1	Angry Recall
Ensemble	0.8413	0.8403	0.8410	0.8029
ResNet50	0.8294	0.8276	0.8287	0.7518
EfficientNetB0	0.8276	0.8280	0.8274	0.7956
Xception	0.8072	0.8048	0.8064	0.6715
InceptionV3	0.7713	0.7707	0.7724	0.7372
MobileNetV2	0.7321	0.7258	0.7279	0.5620
Baseline CNN	0.6195	0.6196	0.6189	0.6788

The ensemble achieved the highest macro F1-score of **0.8410** and macro recall of **0.8403**.

This indicates that the ensemble's improvement was not limited to overall accuracy but extended across the four emotion classes.

15 Final Ensemble: Detailed Class Performance

The detailed classification results for the final ensemble are shown below.

Table 6: Detailed Test Performance of EfficientNetB0 + ResNet50

Class	Precision	Recall	F1-score	Support
Angry	0.83	0.80	0.82	137
Happy	0.85	0.87	0.86	156
Relaxed	0.87	0.85	0.86	139
Sad	0.81	0.84	0.82	154
Macro Average	0.84	0.84	0.84	586

The ensemble performed particularly well on the Happy and Relaxed classes, both achieving an F1-score of 0.86.

The Angry class achieved a recall of 0.80 and F1-score of 0.82, indicating that the model correctly identified approximately four out of every five Angry images.

16 Model-Wise Analysis

16.1 Baseline CNN

The custom CNN achieved a test accuracy of only **61.95%**.

Its macro F1-score was **61.89%**. This establishes a useful baseline but demonstrates that learning visual features entirely from the available dataset was substantially less effective than using pretrained representations.

16.2 MobileNetV2

MobileNetV2 achieved **73.21%** test accuracy and a macro F1-score of **72.79%**.

Although it provided a significant improvement over the baseline, its Angry recall was only **56.20%**.

16.3 EfficientNetB0

EfficientNetB0 achieved **82.76%** test accuracy and **82.74%** macro F1.
Its Angry recall was **79.56%**, while its Relaxed-class recall was particularly strong at 88%.

16.4 ResNet50

ResNet50 achieved the strongest performance among the individual models, with:

- Test accuracy: **82.94%**
- Macro recall: **82.76%**
- Macro F1: **82.87%**
- Angry recall: **75.18%**

It performed particularly well on the Relaxed class.

16.5 Xception

Xception achieved **80.72%** test accuracy and **80.64%** macro F1.
Its Relaxed class performance was strong, but Angry recall was lower at **67.15%**.

16.6 InceptionV3

InceptionV3 achieved **77.13%** test accuracy and **77.24%** macro F1.
Although it performed substantially better than the custom CNN, it was weaker than ResNet50, EfficientNetB0 and Xception.

17 Ensemble Analysis

The final EfficientNetB0 + ResNet50 ensemble achieved:

- Validation accuracy: **85.07%**
- Test accuracy: **84.13%**
- Test macro recall: **84.03%**
- Test macro F1: **84.10%**
- Angry recall: **80.29%**

The strongest individual model, ResNet50, achieved 82.94% test accuracy.
Therefore, the ensemble improved test accuracy by:

$$84.13 - 82.94 = \boxed{1.19 \text{ percentage points}}$$

This suggests that the two networks learned complementary representations.
EfficientNetB0 contributes efficient multi-scale feature extraction, while ResNet50 provides deep residual representations. Combining their feature vectors allows the final classifier to exploit information learned by both architectures.

18 Validation–Test Generalization

The ensemble achieved:

$$Validation\ Accuracy = 85.07\%$$

and:

$$Test\ Accuracy = 84.13\%.$$

The difference is:

$$85.07 - 84.13 = 0.94$$

percentage points.

For ResNet50, the difference was:

$$83.68 - 82.94 = 0.74$$

percentage points.

These relatively small gaps suggest that the final pretrained models generalized reasonably well from validation data to unseen test data.

19 Key Findings

The major findings of the project are:

1. The custom CNN achieved 61.95% test accuracy.
2. Every pretrained architecture substantially outperformed the custom CNN.
3. ResNet50 was the strongest individual model with 82.94% test accuracy.
4. EfficientNetB0 achieved almost identical individual performance at 82.76%.
5. Xception achieved 80.72% test accuracy.
6. InceptionV3 achieved 77.13% test accuracy.
7. MobileNetV2 achieved 73.21% test accuracy.
8. The EfficientNetB0 + ResNet50 ensemble achieved the highest test accuracy of 84.13%.
9. The ensemble also achieved the highest macro F1-score of 84.10%.
10. The ensemble achieved 80.29% Angry-class recall.
11. The ensemble improved upon the strongest individual model by 1.19 percentage points in test accuracy.

20 Limitations

Despite the strong experimental results, several limitations remain.

1. **Subjectivity of emotion labels:** Dog emotion is inherently difficult to define from images alone. Facial appearance may not completely represent the underlying emotional state.
2. **Image-only information:** The models operate on static images and cannot use temporal behavioural information.
3. **Dataset limitations:** Model performance is dependent on the quality, diversity and representativeness of the available images.
4. **Domain generalization:** Performance may decrease on images captured under conditions significantly different from the training dataset.
5. **Interpretability:** CNN predictions do not inherently explain which visual regions influenced the classification.

21 Future Work

Several extensions could improve the project further:

1. Apply Grad-CAM or other explainable AI techniques to identify image regions contributing to emotion predictions.
2. Expand the dataset with greater breed, pose, lighting and background diversity.
3. Investigate video-based emotion recognition using temporal information.
4. Explore attention-based architectures and modern vision transformers.
5. Perform external validation on an independently collected dataset.
6. Investigate class-specific augmentation strategies, particularly for difficult classes such as Angry.
7. Optimize the final ensemble for deployment on resource-constrained devices.

22 Conclusion

This project developed a complete deep learning pipeline for four-class dog emotion classification.

A custom CNN was first established as a baseline and achieved a test accuracy of **61.95%**. Five ImageNet-pretrained CNN architectures were then evaluated using transfer learning and fine-tuning. Every pretrained model substantially outperformed the custom CNN, demonstrating the effectiveness of transfer learning for this task.

Among the individual architectures, ResNet50 achieved the highest test accuracy of **82.94%**, closely followed by EfficientNetB0 at **82.76%**.

These two models were subsequently combined using feature-level ensemble learning. The resulting EfficientNetB0 + ResNet50 ensemble achieved the best overall performance, with:

$$\boxed{\text{84.13\% Test Accuracy}}$$

$$\boxed{84.03\% \text{ Macro Recall}}$$

$$\boxed{84.10\% \text{ Macro F1}}$$

and:

$$\boxed{80.29\% \text{ Angry Recall}}$$

The ensemble improved test accuracy by 1.19 percentage points compared with the strongest individual model.

Overall, the experiments demonstrate that transfer learning combined with careful fine-tuning and feature-level ensemble learning provides a substantially stronger solution than training a CNN from scratch for dog emotion classification.

Final Project Summary

Problem: Four-class dog emotion classification

Classes: Angry, Happy, Relaxed, Sad

Split: 70% Train / 15% Validation / 15% Test

Baseline: Custom CNN – 61.95% test accuracy

Best Individual: ResNet50 – 82.94% test accuracy

Final Model: EfficientNetB0 + ResNet50 Ensemble

Final Accuracy: 84.13%

Macro F1: 84.10%

Angry Recall: 80.29%

End of Report